

Human Position and Angle Detection System

AI/ML Internship — Day 7

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Project Folder: Raritone

Module: Human Body Detection System

1. Introduction

This project focuses on developing an AI-Based Human Position and Angle Detection System using Artificial Intelligence, MediaPipe, OpenCV, and Computer Vision technologies.

The system analyses uploaded human body images and performs:

- body landmark detection
- skeletal visualization
- body posture analysis
- body angle calculation
- body position tracking

The AI model detects body orientation and analyses posture using human pose estimation techniques.

2. Objective

The objective of this project was to:

- detect human body landmarks
 - analyse body posture
 - calculate body angles
 - detect body position
 - improve AI pose estimation systems
 - study skeletal tracking workflows
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3. Technologies Used

Technology	Purpose
Python	Programming

OpenCV	Image Processing
MediaPipe	Human Pose Detection
Artificial Intelligence	Body Analysis
Computer Vision	Landmark Tracking

4. Features Implemented

The AI system successfully implemented:

- body landmark detection
 - skeletal tracking
 - body angle estimation
 - body posture analysis
 - body position detection
 - AI pose visualization
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5. Workflow of the System

Step 1 — Human Image Input

The AI system accepts uploaded human body images for analysis.

Step 2 — Body Landmark Detection

MediaPipe detects:

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

using AI pose estimation.

Step 3 — Coordinate Extraction

The AI system extracts:

- x coordinates
- y coordinates

for body landmark analysis.

Step 4 — Body Angle Calculation

The body angle is calculated using landmark coordinates.

The distance and angle calculations help analyse:

- body tilt
- arm orientation
- body posture
- shoulder alignment

The angle calculation uses:

$$\theta = \tan^{-1} \left(\frac{y_2 - y_1}{x_2 - x_1} \right)$$

-10-8-6-4-2246810-10-5510A(-8, -8)B(8, 8)m = 1.00

Step 5 — Position Detection

The AI system detects:

- standing straight posture
- tilted body posture
- body orientation
- movement direction

using landmark alignment.

6. Code Implementation:

```
1  √ import cv2
2    import mediapipe as mp
3    import math
4
5    image = cv2.imread("woman.jpg")
6
7  √ if image is None:
8      print("Image not found")
9      exit()
10
11   image = cv2.resize(image, (720, 720))
12
13   mp_pose = mp.solutions.pose
14   mp_draw = mp.solutions.drawing_utils
15
16  √ pose = mp_pose.Pose(
17      static_image_mode=True,
18      min_detection_confidence=0.7
19  )
20
21   rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
22
23   results = pose.process(rgb)
24
25  √ if results.pose_landmarks:
```

```
26
27     lm = results.pose_landmarks.landmark
28
29     h, w, _ = image.shape
30
31     ls = lm[11]
32     rs = lm[12]
33
34     x1, y1 = int(ls.x*w), int(ls.y*h)
35     x2, y2 = int(rs.x*w), int(rs.y*h)
36
37     angle = int(
38         math.degrees(
39             math.atan2(y2-y1, x2-x1)
40         )
41     )
42
43     if abs(y1-y2) < 20:
44         position = "Standing Straight"
45
46     elif y1 > y2:
47         position = "Tilted Right"
48
49     else:
50         position = "Tilted Left"
```

```

51
52     mp_draw.draw_landmarks(
53         image,
54         results.pose_landmarks,
55         mp_pose.POSE_CONNECTIONS
56     )
57
58     cv2.putText(
59         image,
60         f"Angle: {angle}",
61         (20,40),
62         cv2.FONT_HERSHEY_SIMPLEX,
63         0.8,
64         (0,255,255),
65         2
66     )
67
68     cv2.putText(
69         image,
70         f"Position: {position}",
71         (20,90),
72         cv2.FONT_HERSHEY_SIMPLEX,
73         0.8,
74         (0,255,0),
75         2
76     )

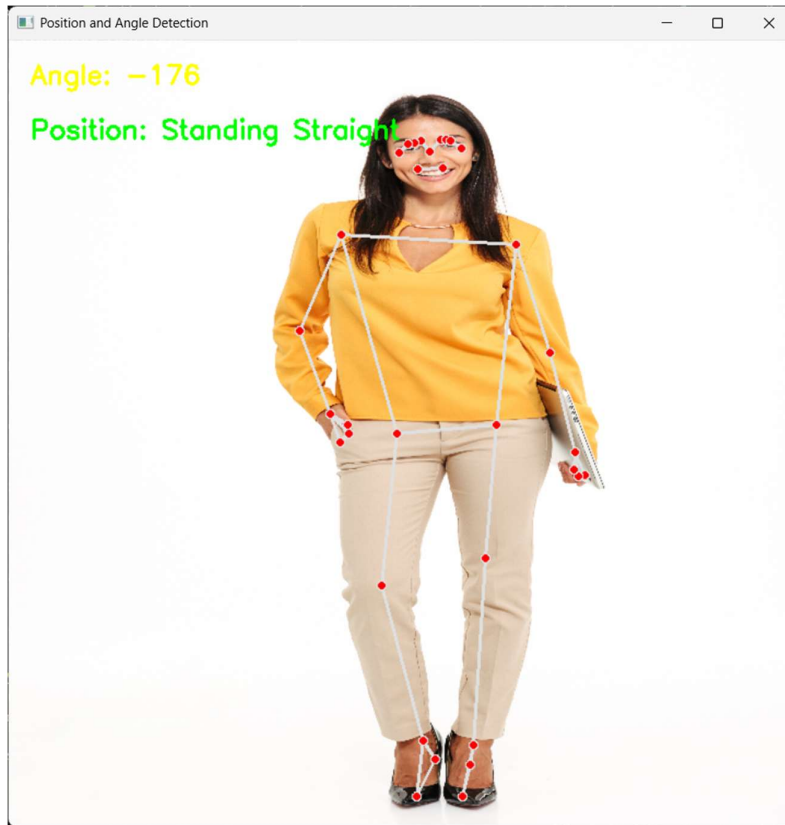
```

```

75         2
76     )
77
78     print("Angle:", angle)
79     print("Position:", position)
80
81 else:
82     print("No Human Detected")
83
84 cv2.imwrite("output.jpg", image)
85
86 cv2.imshow("Position and Angle Detection", image)
87
88 cv2.waitKey(0)
89
90 cv2.destroyAllWindows()

```

7. Output:



8. Applications

The AI system can be used in:

- fitness monitoring
- posture correction systems
- sports analysis
- rehabilitation systems
- AI motion tracking
- healthcare applications

9. Advantages

The advantages include:

- real-time posture analysis
- accurate body angle detection

- AI skeletal tracking
 - intelligent body movement analysis
 - improved pose estimation
-

10. Challenges Faced

Challenge	Impact
Poor lighting	Reduces landmark visibility
Fast movement	Reduces tracking stability
Side body posture	Affects angle accuracy
Complex background	Reduces detection quality

11. Expected Output

The AI system generates:

- body landmark visualization
 - skeletal tracking
 - body angle analysis
 - posture detection
 - body position estimation
-

12. Deliverables

The project deliverables include:

- AI pose detection system
 - body angle detection
 - body posture analysis
 - skeletal visualization
 - AI testing screenshots
 - output image generation
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13. Learning Outcomes

During this project, the following concepts were understood:

- human pose estimation
 - body landmark tracking
 - skeletal visualization
 - body posture analysis
 - AI body angle estimation
 - computer vision workflows
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14. Conclusion

The AI-Based Human Position and Angle Detection System successfully demonstrated:

- AI pose estimation
- body posture analysis
- skeletal tracking
- body angle detection
- intelligent body position analysis

using Artificial Intelligence and Computer Vision technologies.

The project improved understanding of:

- human landmark detection
- body angle calculation
- posture analysis systems
- AI skeletal tracking
- computer vision applications